

The Institutional Flow Clock

A pre-registered test of pre-positioning for calendar-scheduled institutional flows

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Code, data scripts and full audit trail: <https://github.com/minh-stakc/gqh-flow-clock>

1. Summary

Pension, target-date and balanced funds, index-benchmarked bond managers and institutions raising month-end cash must trade on calendars fixed by mandates, index rules and settlement plumbing, and they pay for immediacy when they do. We take the position two to four days ahead of these flows and hand it to them at the month-end close. **New here:** one “flow clock” combining three such flows; a natural experiment that moves the dash-for-cash window with each US settlement change (T+3, T+2, T+1); sizing the Treasury index-extension trade by the month’s announced coupon issuance; and a volatility-invariant path statistic for trend. All were pre-registered before any backtest (commit d20c9d5); the last three failed in-sample. A logged, overfitting-tested search (1,404 configurations, probability of backtest overfitting 0.023) selected **rebalancing pressure (A) + Treasury month-end index extension (C) + time-series momentum**: in-sample (2006–2024) Sharpe **0.98** net of costs (0.62 at 2x costs), maximum drawdown -11.7% , factor alpha 6.2% a year ($t = 3.95$), but a Deflated Sharpe of only 0.40 under the pre-registered trial count. **The single out-of-sample run (Oct 2024–Oct 2026) returned a Sharpe of -0.12** (-0.61 at 2x costs, 0.38 before costs): 3.0% a year, below T-bills. Both flow sleeves lost money (A even before costs); only the trend leg earned. The same rules on CME futures, at about a third of the trading cost, earned 1.05 in-sample and 0.01 out-of-sample.

2. Economic hypothesis: who is on the other side?

Sleeve	Forced trader and why they cannot stop	Rule (decide at close, fill next close)	Prediction / kill condition
A. Rebalancing pressure	60/40, target-date and pension funds must sell the asset that outperformed back to fixed weights at month end (Harvey, Mazzoleni & Melone 2025); policy is set by committees	Drift of a 60/40 SPY/IEF portfolio at $T-6$, standardised; hold $-$ drift in SPY vs IEF over $[T-4, T]$, 5% vol per leg	SPY–IEF over the window opposes the drift
B. Dash for cash	Institutions must sell to have cash on payment dates; the last trade that settles by month end is $T-k$ (Etula et al. 2020)	Short SPY over $[T-k-5, T-k-1]$, long over $[T-k, T+3]$; $k = 3/2/1$ under T+3/T+2/T+1	<i>New:</i> windows shift with each settlement change
C. Treasury index extension	Bond indices add new issues only at month end; benchmarked managers must buy duration into it (Hartley & Schwarz 2019)	Long IEF over $[T-2, T]$, sized by the month’s announced DV01 of coupon issuance (<i>new</i>)	IEF excess return over the window > 0
Trend leg	Under-reaction to slow information (Moskowitz, Ooi & Pedersen 2012)	12-month sign, vol-scaled, 15 ETFs	Diversifies the flow sleeves

Who pays us, and why it is not arbitrated away. We are not the flow’s counterparty at entry: we warehouse the position for two to four days and become its counterparty at the month-end

close, when a fund that must trade a fixed amount at a fixed time pays for immediacy (Figure 2). These funds are price-insensitive by mandate: changing a rebalancing policy, an index rule or the settlement cycle takes governance, not a trading decision; arbitrage capital stays limited because each window is a short, levered bet whose expected gain is small relative to its risk. It dies if the patterns are crowded after publication, the flows move earlier, or costs exceed the effect. The trend variant PCT (extending the author’s earlier work) scales positions by $R = A/(L\sqrt{W})$, mean absolute deviation from the chord over mean absolute daily return ($\pi/8$ under a random walk).

3. Data and universe

Total-return-adjusted daily ETF prices (Yahoo Finance), the 3-month T-bill (FRED DTB3), Fama-French 5 factors and momentum (K. French library, to Aug 2026), and every Treasury auction since 2002 with its announcement date (U.S. Treasury Fiscal Data). Replication: Databento GLBX.MDP3 CME futures (20 markets, including ES and ZN), volume-ranked front contract, returns always within one contract plus one round trip per roll; ES/ZN sampled at 16:00 ET to match ETF closes (daily correlation with SPY/IEF 0.998/0.987). History 2005-01 to 2026-10-02; the brief’s 20%-or-2-years rule puts the **out-of-sample** window at 2024-10-03 to 2026-10-02 (501 days, 24 month ends); with data from 2017 the 20% rule binds instead (out-of-sample from 2024-10-18) and the same strategy gives 0.93 in-sample and -0.09 out-of-sample. *Survivorship*: broad ETFs that still trade (a mild survivor choice; DBC, SLV, UUP, HYG launched in 2006–07 and enter after 300 days) and futures; *corporate actions*: total-return adjustment; *missing data*: none after each fund’s first print. An audit found that counting window days on the realised calendar leaked unscheduled closures (Hurricane Sandy, 2012) before they were announced; we fixed it (sleeve C’s Sharpe moved from 0.78 to 0.77).

4. Methodology

Protocol and audit trail (git commit, UTC, 2026-10-03). (i) d20c9d5 at 10:45: hypotheses, rules, variant neighbourhoods, selection rule, gates and the out-of-sample data lock; the first backtest in `results/trials.csv` is at 10:53. (ii) 06b22c9 at 11:02: Amendment 1 (ensemble search with overfitting controls). By then build agents had logged 89 sleeve-level in-sample runs (33 specifications), which the author had not opened; no ensemble had been run. (iii) b054990 at 11:45: `results/selection.json` frozen with the trial log, which holds no out-of-sample row. (iv) 11:46: the out-of-sample window, which the loaders refuse to return unless unlocked, evaluated once (every row of `results/oos_log.csv` carries that minute; commit messages were later edited, contents and dates unchanged, and `results/commit_map.csv` maps the hashes in the logs). The log holds 3,134 in-sample runs, of which 1,467 are distinct specifications at 1x costs (the rest are 2x-cost repeats and post-fix re-runs); 1,467 enters the Deflated Sharpe.

Execution and costs. Signals use data up to the close of day d and fill at the close of $d+1$ (ETF trend: next open). One-way costs: 3 bp for SPY/IEF-class ETFs (several times the half-spread), 5 bp sector/commodity/currency ETFs, 10 bp otherwise, 30 bp/yr to borrow shorts; futures 1 bp (ES, ZN; only about 1.4 half-ticks) to 5 bp; cash earns the T-bill; every result is repeated at 2x costs.

Ensemble and selection. Streams: sleeves A–E (D: FX hedge flows, Melvin & Prins 2015; E: auction cycle, Lou, Yan & Zhang 2013; both exploratory) and a trend leg (PCT or plain TSMOM). The search spans stream subsets, three weightings (equal risk, shrunk inverse volatility, Ledoit-Wolf 2004 minimum variance; expanding windows, lagged), volatility targets of 6/8/10% and an optional drawdown brake. Rule: highest in-sample Deflated Sharpe ratio (Bailey & López de Prado 2014)

among configurations passing G1 (Sharpe > 0 at 2x costs), G2 (no year $> 40\%$ of P&L) and G3 (median Sharpe of one-step neighbours $\geq$ half its own), provided the CSCV probability of backtest overfitting (Bailey et al. 2017; 16 blocks, 12,870 splits) is below 0.25.

Verification. Each module was implemented by an AI coding agent from the pre-registered text and attacked by a separate AI reviewer for lookahead (random-walk perturbation, `tests/test_lookahead.py`), spec drift and bugs, re-deriving headline numbers with separate code. The organisers’ backtrader starter matches our engine on identical weights (mean gap 0.0006 bp a day); as one netted book at the guide’s example 5+5 bp per side it gives 0.86 in-sample and -0.19 out-of-sample, and its $r_f=0$ Sharpe reads 1.17/0.48 for our returns (`validation/`). Two bugs were fixed before selection: the closure-calendar leak and an overlay that charged re-scaling costs on idle streams (an impossible -124% day); the search was re-run. A clean clone with freshly downloaded data reproduces every number here.

5. Results

Table 1: Submitted strategy (A + C + TSMOM, minimum-variance weights, 8% vol target), net of costs except “Gross”. Ann. ret is total return including T-bill interest (T-bill 1.6% in-sample, 4.0% out-of-sample); all Sharpe ratios use returns over the T-bill, hence a positive return with a negative Sharpe. Turnover: all streams’ trades / NAV per year. NW t : Newey-West t of mean daily excess return.

	Ann. ret	Vol	Sharpe	2x costs	Gross	Max DD	Worst mo.	Turnover	NW t
In-sample 2006–24	9.0%	7.6%	0.98	0.62	1.35	−11.7%	−4.7%	90×	4.62
Out-of-sample 2024–26	3.0%	6.6%	−0.12	−0.61	0.38	−8.7%	−2.3%	108×	−0.18

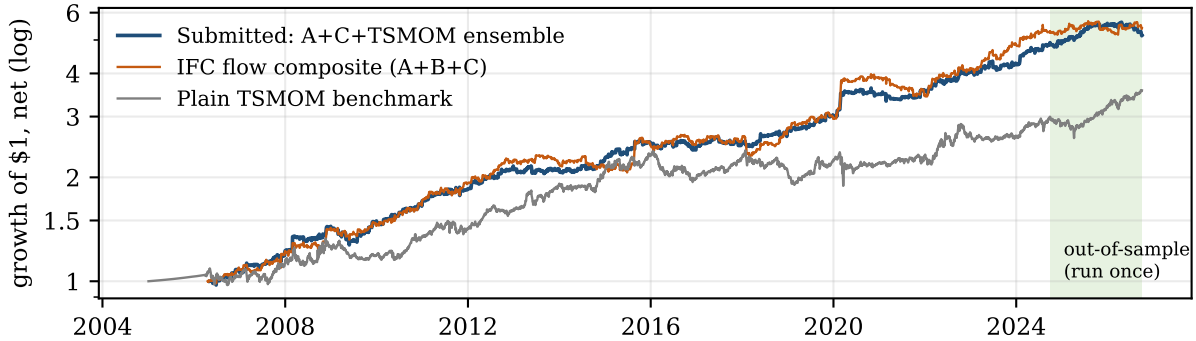


Figure 1: Growth of \$1, net of costs. The shaded period was evaluated once, after the selection was committed.

In-sample evidence. Sub-period Sharpe 1.29/1.23/0.57/1.09/0.79 (2006–08, ..., 2021–24); best year 11% of P&L; 90% block-bootstrap interval [0.67, 1.31]. Factor loadings are small but non-zero: market 0.05 ($t = 3.6$), momentum 0.05 ($t = 5.0$), Treasury duration (IEF excess) 0.31 ($t = 9.3$, from sleeve C); $R^2 = 0.09$. The pick ranks first of 1,404 configurations (median 0.33) and its one-step neighbours have median Sharpe 0.94, so it is not a knife edge; but in the CSCV the training winner’s Sharpe fell from 1.07 to 0.75 on the held-out half, a selection haircut of about 30%. On the pre-registered count of 1,467 trials the best unskilled trial is expected to reach 1.04, above our 0.98, so the Deflated Sharpe is only 0.40: after the search, the in-sample Sharpe is not statistically significant. With the post-hoc effective count (3.5, eigenvalue participation, likely too low) it is 0.999. Timing matters: shifting every flow window a day earlier/later gives 0.52/0.53 (A+B+C composite), one extra day of delay 0.59.

Table 2: Every pre-registered stream and replication: net Sharpe ratio. Futures statistics start 2011-09 after warm-up.

ETF streams	IS	OOS	OOS 2x	Composites / futures	IS	OOS
A rebalancing pressure	0.47	-0.73	-1.01	IFC flow composite A+B+C	0.94	-0.31
B dash for cash	0.19	-0.27	-0.42	IFC-F on ES/ZN (Databento)	1.05	0.01
C Treasury month end	0.77	-0.09	-0.45	PCT-F, 20 futures	0.24	-0.10
D FX hedge (exploratory)	-0.68	0.28	-0.37	TSMOM-F, 20 futures	0.28	0.14
E auction cycle (exploratory)	0.16	-1.06	-1.79	ID-conditioned trend-F	0.43	0.06
PCT (path-geometry trend)	0.30	0.71	0.60			
TSMOM benchmark	0.44	0.55	0.49			
ID-conditioned trend	0.57	0.49	0.43			

What failed in-sample. The settlement natural experiment: fixed T+3 windows beat the settlement-aware ones (0.23 vs 0.19; 0.42 vs 0.30 within the T+2 regime), so B2 is rejected. Issuance sizing added nothing to C (0.78 without it). PCT did *not* beat plain TSMOM (0.30 vs 0.44; futures 0.24 vs 0.28), so its kill condition fired. The information-discreteness variant (Da, Gurun & Warachka 2014) beat TSMOM in-sample on both universes (0.57, 0.43), though the ETF difference’s 90% interval $[-0.01, 0.29]$ includes zero. D and E failed (E is negative at 2x costs).

Out-of-sample. The gross Sharpe fell from 1.35 to 0.38: the signal weakened, and costs on $108\times$ turnover, which the in-sample edge had absorbed, made it negative. Contributions in-sample $\rightarrow$ out-of-sample: A 2.2% $\rightarrow$ -1.3%, C 4.4% $\rightarrow$ -0.2%, trend 1.2% $\rightarrow$ 1.1% a year; factor alpha -1.3% ($t = -0.35$). Implied gross Sharpe: A -0.46, C 0.27. By year the Sharpe was 0.33 (Q4 2024), 1.12 (2025, in line with history) and -1.99 (2026 to date, the worst year on record; worst full in-sample year -0.22): in 2026 A and C each lost about 5% (-5.4%, -5.0%) while trend earned 1.6%. This haircut matches the median backtest-to-live haircut of 215 bank risk-premia indices (1.20 $\rightarrow$ 0.31; Suhonen, Lennkh & Perez 2017). Kill conditions did not trigger (Figure 2): IEF earned 3.0 bp/day over $[T-2, T]$ versus -1.6 on other days from $T-10$ to $T+5$ (in-sample 7.2 vs -0.4), and the drift-signed bond-minus-stock spread over $[T-4, T]$ stayed positive but fell from 11.0 to 2.5 bp/day. The settlement experiment, now entirely in T+1, failed again (B -0.27). Twenty-four month ends cannot separate three explanations: decay (the rebalancing paper appeared in early 2025, yet A earned 1.1% in 2025 and broke only in 2026; McLean & Pontiff 2016), selection (Deflated Sharpe 0.40), and noise (a two-year Sharpe has a standard error near 0.7, though the result is in the bottom 1% of the strategy’s overlapping in-sample two-year windows, median 0.97, minimum -0.29). PCT was the best out-of-sample ETF stream (0.71) but the worst futures trend variant (-0.10); switching would be selection on the test set.

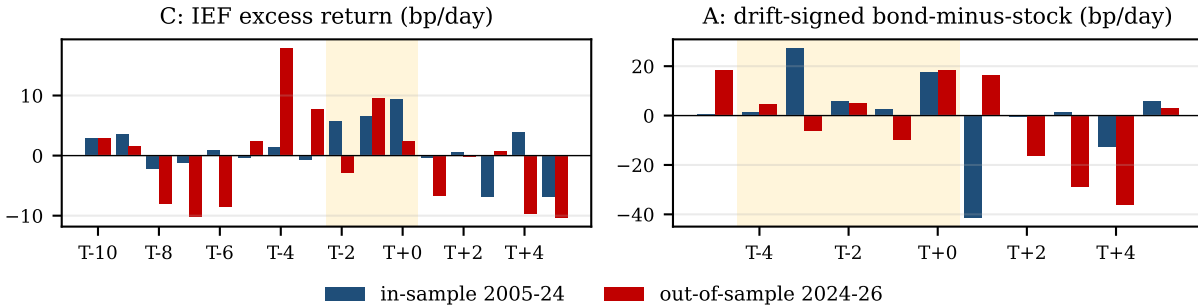


Figure 2: Mean daily excess return by trading day relative to month end T , no strategy code; shaded: traded window. Right: bond-minus-stock spread signed by the 60/40 drift observed at $T-6$ (days after T signed by the month just ended); in-sample it reverses on $T+1$ (-42 bp), as temporary price pressure predicts.

6. Risk management

Limits. One hard cap, gross exposure $\leq 4 \times$ NAV summed over streams, which is also the effective per-name limit: IEF reaches it in Treasury month-end windows (daily gross averages $0.87 \times$, 95th percentile $3.88 \times$; the $3 \times$ per-ETF cap applies to the IFC composite only). The cap bounds a 3-day $4 \times$ IEF window at about 3.0% per one-sigma move (6.0% at two sigma); the worst in-sample month was -4.7% . **Pre-set de-risking.** Built in and tested: per-stream volatility scaling and an 8% portfolio volatility target (lagged 63-day EWMA), so exposure falls after volatility spikes. The pre-registered drawdown brake (halve exposure while more than 10% below peak) was searched but not selected: here it fired once, in mid-2009, and lowered the Sharpe from 0.98 to 0.97. *Proposed after the out-of-sample run, untested:* suspend a sleeve when its kill statistic (Figure 2) is negative over 24 months, and skip windows containing CPI, payrolls or FOMC releases. **Factor, regime and tail.** Duration is the main macro risk (a long-duration bet three days a month). Out-of-sample the market beta rose to 0.13 ($t = 7.1$) and the duration beta to 0.46 ($t = 5.0$): the book was more directional than designed. The strategy made money in 2008 and 2020 (flight to quality) and in 2022, when stocks and bonds fell together. Daily skew is positive (1.04), so the flow legs are not short volatility, but returns come in jumps: daily kurtosis 24.7, best month 12.2% (February 2020).

7. Liquidity and capacity

Square-root impact (Almgren et al. 2005): cost per trade = fixed + $1.0 \times \sigma_{\text{daily}} \sqrt{\text{trade}/\text{ADV}}$, using trailing 63-day median dollar volume and the strategy’s actual trades, starting from gross returns. Net Sharpe by assets under management:

AUM	\$1k (fixed costs only)	\$1M	\$10M	\$100M	\$1B
ETF ensemble (2019–24 liquidity)	1.04	0.82	0.31	−1.15	−3.47
Futures A+B+C, IFC-F (2011–24)	0.99	0.98	0.94	0.83	0.47

The ETF version concentrates up to $4 \times$ NAV of IEF (median volume \$0.6bn a day in 2019–24) into month-end windows, so it is an 18-year test of the effect, not a vehicle: its edge halves near \$5M. ZN and ES trade about \$171bn and \$308bn a day, so the futures flow composite (not the exact submitted mix) keeps half its edge near \$1bn (in-sample 0.92 at 2x costs). At $90 \times$ turnover a year the edge must stay large; out-of-sample it did not, even in futures (0.01).

8. Limitations, deviations and next steps

A and C come from papers whose samples overlap ours, so the out-of-sample window was the only independent test, and the strategy failed it. **Deviations from the pre-registration:** Amendment 1 replaced the selection rule (the original rule would have submitted the IFC composite, -0.31 out-of-sample); the pre-registered one-day-delay and window-shift variants were not re-run out-of-sample; the effective-trial DSR, the rolling-window comparison, the capacity window and the out-of-sample diagnostics are post hoc. **Next:** do not trade this as is. Before any of their data exists we froze two forward tests (FORWARD_TEST.md, FORWARD_TEST_2.md, git-tagged): the submission unchanged, its CME-futures version, the information-discreteness trend, and the three strategies with the best live evidence in the literature (broad CME trend, equities plus trend, Faber’s GTAA), expected Sharpe 0.4–0.6, not 1. History computed only after the freeze (in/out of sample): futures version 0.96/−0.10, trend $-0.13/0.14$, equities plus trend 0.50/0.46, GTAA 0.51/0.70. Only returns from 2026-10-06 count; quarterly reports are published whatever they show.

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Data. Yahoo Finance (via `yfinance`); FRED, Federal Reserve Bank of St. Louis; Kenneth R. French Data Library; U.S. Treasury Fiscal Data (auctions); Databento GLBX.MDP3. **Prior work extended:** github.com/minh-stakc/physics-quant-research. **AI use:** code was written and audited with AI coding agents under the author’s direction (see Methodology); the hypotheses, pre-registration, selection rule and this note are the author’s responsibility. **Reproduce:** `python reproduce.py` (see README).